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The latter question is particularly important because numerical stress-intensity factors are not merely outputs of an incremental calculation: they determine the next geometry and therefore influence all subsequent solutions. Interaction integrals offer a means of separating the mixed-mode contributions [15], while displacement-extrapolation estimates are sensitive to near-tip element arrangements [16]. Comparisons of extraction procedures have emphasized the advantages of integral methods [17] and their dependence on adequate finite-element resolution [18]. Mixed-mode assessments have likewise examined the response of interaction-integral estimates to discretization choices [19]. For a recursively generated path, such local extraction differences can also alter the later crack geometries. Accuracy checks for a prescribed tip must therefore be distinguished from comparisons of independently propagated trajectories.

Here we examine this distinction for a single crack initiating at an asymmetric circular hole in a finite plate. The maximum tensile tangential stress of the uncracked-hole solution determines the initiation point and its outward material-side normal. Only the first crack segment is prescribed along that normal. At each subsequent step, the current elastic boundary-value problem is solved for the complete accumulated crack geometry, and the next direction is selected from the current-tip SIFs using the MTS criterion. The objective is not to introduce a new fracture criterion, but to establish which features of the resulting numerical trajectory remain stable under the investigated changes in extraction and discretization.

The assessment combines prescribed-field and fixed-geometry checks with independent crack-path calculations. Interaction-integral SIFs are compared with crack-opening-displacement (COD) extrapolations from the same computed displacement field; a centered-hole control probes reflection symmetry and the local one-step angular response. Independently propagated paths then test changes in exterior-mesh density and structured-tip resolution. A separate family with 4-, 2-, and 1-mm increments also refines the numerical length scales, so that it examines coupled increment/discretization sensitivity rather than the increment in isolation. We examine the positive maximum and subsequent sign reversal of mode mixity, their effect on the MTS turning direction, and the behavior of increment-normalized turning measures under coupled refinement.

Sections 2 and 3 define the physical problem and the incremental MTS construction. Sections 4 and 5 present the numerical assessment and crack-trajectory results, followed by the physical interpretation, limitations, and conclusions in Sections 6 and 7.

2 Problem formulation

2.1 Geometry, material, and loading

Figure 1 introduces the plate geometry, the coordinate conventions, and the applied loading. The specimen has width $2A$ and height B, with a circular hole of radius R whose center is denoted in the schematic by (x_c, y_c), where x_c and y_c are the coordinates measured from the lower-left corner. The computational Cartesian frame has its origin at the midpoint of the left edge, with

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Two exterior mesh-sizing laws are considered. The reference law, denoted M0, uses a far-field element-size cap of 2.5 mm, a grading slope of 0.10, and a boundary-metric growth parameter of 0.25. The coarser law M1 increases these values to 5.0 mm, 0.15, and 0.35, respectively. Both laws retain the 4-mm grading transition length in the reference-increment comparisons. M0 is used for the reference 4-mm trajectory and the structured-tip resolution tests, including the independently propagated $2h_0$ path. M1 is used for the fixed-geometry exterior-mesh comparison and the independently propagated exterior-mesh trajectory. The coupled 4/2/1-mm refinement study, described below, uses dimensionless versions of the M1 exterior law and a coarser structured-tip family.

For the fixed P_{17} and P_{22} geometries, changing from M0 to M1 reduces the total T3 element count by approximately 31–33% and the exterior count by 40–43%. The paired 12678-element near-tip core and its 11316-element interaction-domain support remain unchanged, permitting a comparison of the extracted SIFs before the M1 trajectory is propagated independently.

4.3 Structured-tip resolution tests

The influence of structured-tip resolution is first examined on fixed accepted crack geometries. The mesh spacing near the current tip is varied while the complete crack polyline, material, loading, interaction-domain radii, far-field cap, transition length, and MTS criterion remain unchanged. The production tip size is denoted by $h_0 = 0.027012$ mm. The three physically compared resolutions are

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These correspond to structured-core scale factors 2, 1, and 1/2. The paired-core radius remains $r_{\text{core}} = 3$ mm and the EDI annulus remains $r_{\text{in}} = 0.4$ mm to $r_{\text{out}} = 2.6$ mm. The paired cores contain 3318, 12678, and 49518 T3 elements, respectively. COD fitting uses native crack-face node pairs, namely corresponding upper and lower T6 nodes at the same distance behind the crack tip. Their sampling adequacy is discussed with the extraction procedure in Section 4.5.

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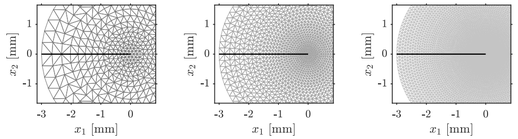
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(a) $4h_0$ (b) $2h_0$ (c) h_0

Figure 2: Structured-tip meshes with a common core radius $r_{\mathrm{core}} = 3$ mm. The reference spacing is $h_0 = 0.027012$ mm; the illustrative $4h_0$ level does not satisfy the COD sampling criterion.

$(\boldsymbol{\sigma}, \boldsymbol{\varepsilon}, \boldsymbol{u})$, let Ω_k denote the integration domain around the current tip. Here $\mathrm{d}\Omega$ is the area element, a comma followed by an index denotes partial differentiation in the local frame, and the colon denotes tensor contraction. The interaction integral reads

$$I^{(m)} = \int_{\Omega_k} \left[-(\boldsymbol{\sigma}^{(m)} : \boldsymbol{\varepsilon})\delta_{1j} + \sigma_{ij} u^{(m)}_{i,1} + \sigma^{(m)}_{ij} u_{i,1} \right] w_{,j} \, \mathrm{d}\Omega, \qquad i, j \in \{1, 2\}. \tag{20}$$

The gradients of the actual displacement field are evaluated with T6 shape functions, whereas the auxiliary stresses are taken from the asymptotic crack-tip fields. Derivatives of the auxiliary displacements are evaluated by centered finite differences with step $h = \max(10^{-7}\ \mathrm{m}, 10^{-5} r)$, where r is the distance from the current tip. The nodal weight w equals one for $r \le r_{\mathrm{in}}$, decreases linearly to zero at r_{out}, and vanishes beyond the annulus. Its T6 interpolant provides the gradient used in Eq. (20); numerical integration employs the 16-point triangular rule. For unit auxiliary intensity $K_{\mathrm{aux}} = 1$ in plane strain,

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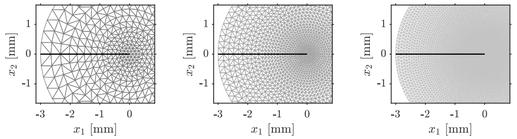
4.5 COD-based extraction comparison

Crack-opening displacement (COD) provides a second route to stress-intensity factors from the same computed elastic field. Each native crack-face node pair consists of matching upper and lower T6 nodes on the last segment, with both nodes located at distance r behind the current tip. The displacement jump $[\![u_i]\!]$ is the upper-minus-lower difference in component u_i at such a pair. These existing nodal positions are used directly; no additional interpolation points are introduced. The corresponding apparent COD SIFs are

$$\begin{bmatrix} K_{\mathrm{I}}^{\mathrm{app}}(r) \\ K_{\mathrm{II}}^{\mathrm{app}}(r) \end{bmatrix} = \frac{\mu}{\kappa + 1} \sqrt{\frac{2\pi}{r}} \begin{bmatrix} [\![u_2]\!] \\ [\![u_1]\!] \end{bmatrix}, \qquad \kappa = 3 - 4\nu. \tag{22}$$

Here κ is the dimensionless plane-strain Kolosov constant. Linear and quadratic fits of the apparent SIFs against the normalized distance $r/\Delta a$ are extrapolated to $r = 0$. The four fitting intervals are $[0.04, 0.20]$, $[0.04, 0.30]$, $[0.08, 0.30]$, and $[0.12, 0.30]$. The intervals overlap, so their sampling counts are not additive. In the stated order, the numbers of native crack-face node pairs are 19, 28, 23, and 18 for the $2h_0$ mesh; 38, 55, 44, and 34 for the reference h_0 mesh; and 74, 108, 86, and 67 for the $h_0/2$ mesh.

The prescribed qualification criterion requires at least 12 pairs in each fitting interval. The illustrative $4h_0$ core provides 10, 15, 12, and 9 pairs, respectively, and fails this criterion in the first and fourth intervals. It is therefore excluded from the physical SIF and MTS comparisons. Linear and quadratic fitting over the four intervals gives eight COD estimates from the same finite-element displacement field, not eight independent elastic solutions. Only the EDI SIFs determine subsequent MTS directions, and the differences among COD fits do not represent a statistical uncertainty interval.



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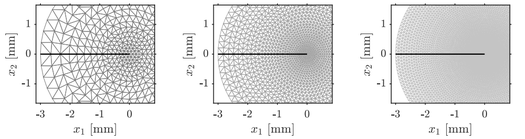
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$$K_m = \frac{E'}{2K_{\mathrm{aux}}} I^{(m)}, \qquad E' = \frac{E}{1 - \nu^2}, \qquad m \in \{I, II\}. \tag{21}$$

4.5 COD-based extraction comparison

Crack-opening displacement (COD) provides a second route to stress-intensity factors from the same computed elastic field. Each native crack-face node pair consists of matching upper and lower T6 nodes on the last segment, with both nodes located at distance r behind the current tip. The displacement jump $[\![u_i]\!]$ is the upper-minus-lower difference in component u_i at such a pair. These existing nodal positions are used directly; no additional interpolation points are introduced. The corresponding apparent COD SIFs are

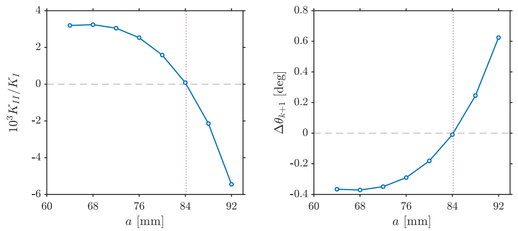
$$\begin{bmatrix} K_{\mathrm{I}}^{\mathrm{app}}(r) \\ K_{\mathrm{II}}^{\mathrm{app}}(r) \end{bmatrix} = \frac{\mu}{\kappa + 1} \sqrt{\frac{2\pi}{r}} \begin{bmatrix} [\![u_2]\!] \\ [\![u_1]\!] \end{bmatrix}, \qquad \kappa = 3 - 4\nu. \tag{22}$$

Here κ is the dimensionless plane-strain Kolosov constant. Linear and quadratic fits of the apparent SIFs against the normalized distance $r/\Delta a$ are extrapolated to $r = 0$. The four fitting intervals are $[0.04, 0.20]$, $[0.04, 0.30]$, $[0.08, 0.30]$, and $[0.12, 0.30]$. The intervals overlap, so their sampling counts are not additive. In the stated order, the numbers of native crack-face node pairs are 19, 28, 23, and 18 for the $2h_0$ mesh; 38, 55, 44, and 34 for the reference h_0 mesh; and 74, 108, 86, and 67 for the $h_0/2$ mesh.

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Table 2: Characteristic accepted states. SIFs have units MPa$\sqrt{\text{m}}$; angles are in degrees. The interpolated row contains only a graphical zero estimate, not a solved stress field or crack direction.

State	a [mm]	K_{I}	K_{II}	$K_{\mathrm{II}}/K_{\mathrm{I}}$	θ_k	$\Delta\theta_{k+1}$
P_1	4	0.36648	4.198×10^{-6}	1.146×10^{-5}	0.0000	-0.00131
P_{17}	68	0.78679	2.548×10^{-3}	3.239×10^{-3}	-2.4444	-0.37115
P_{21}	84	0.96631	7.860×10^{-5}	8.134×10^{-5}	-3.6366	-0.00932
P_{22}	88	1.0667	-2.284×10^{-3}	-2.142×10^{-3}	-3.6460	+0.24541
P_{23}	92	1.2444	-6.780×10^{-3}	-5.449×10^{-3}	-3.4006	+0.62435
Interpolated	84.15	—	—	0	—	—



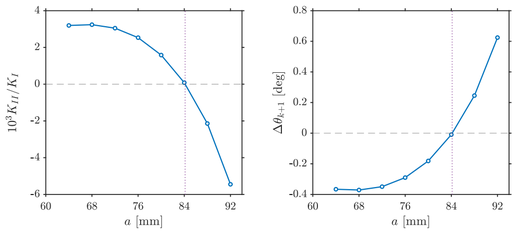
(a) Late mode-mixity history.

(b) Incremental MTS turn.

Figure 5: Mode-mixity sign change and reversal of the MTS turn. The dotted line denotes the interpolated mode-mixity zero. The turn computed at P_{23} ($a = 92$ mm) predicts the proposed, physically unsolved next segment.

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5.3 Mode-mixity sign change and turning reversal

The mode-mixity ratio remains positive at P_{21} ($a = 84$ mm), where $q_{\mathrm{K}}^{(21)} = +8.134267 \times 10^{-5}$, but is negative at P_{22} ($a = 88$ mm), where $q_{\mathrm{K}}^{(22)} = -0.0021416$. Thus the sign reversal is established between two independently solved accepted states, although neither has zero mode mixity. For the graphical marker in Figs. 3b and 5, linear interpolation between their values gives

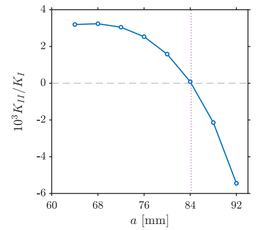
$$a_{\mathrm{LS}} = a_{21} - \frac{q_{\mathrm{K}}^{(21)}}{q_{\mathrm{K}}^{(22)} - q_{\mathrm{K}}^{(21)}}(a_{22} - a_{21}) \simeq 84.15 \text{ mm}. \tag{27}$$

Here a_{LS} denotes only an interpolated estimate of the crack length at which local symmetry ($K_{\mathrm{II}} = 0$) would occur. It is not a solved zero-mode-II state, and the interpolation does not establish the physical location with sub-increment accuracy.

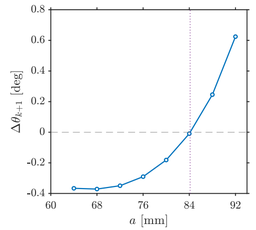
The sign change has an immediate directional consequence. At P_{21} the MTS turn toward P_{22} is still clockwise, whereas the negative mode mixity at P_{22} produces a counterclockwise turn toward the accepted P_{23} segment (Fig. 5). At that last solved state, $K_{\mathrm{I}} = 1.2444$, $K_{\mathrm{II}} = -0.0067803$, and $q_{\mathrm{K}} = -0.0054488$; the MTS rule then predicts $\Delta\theta_{24} = 0.62435°$ and $\theta_{24} = -2.7762°$. This final prediction specifies the qualified extension, not an accepted elastic solution at P_{24}.

Table 2: Characteristic accepted states. SIFs have units MPa$\sqrt{\text{m}}$; angles are in degrees. The interpolated row contains only a graphical zero estimate, not a solved stress field or crack direction.

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Interpolated	84.15	—	—	0	—	—



(a) Late mode-mixity history.



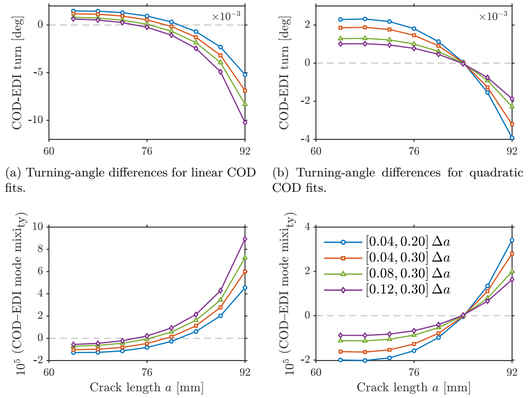
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Figure 5: Mode-mixity sign change and reversal of the MTS turn. The dotted line denotes the interpolated mode-mixity zero. The turn computed at P_{23} ($a = 92$ mm) predicts the proposed, physically unsolved next segment.

5.4 COD comparison and extraction sensitivity

Because the sign reversal of mode mixity changes the MTS turning direction, its persistence under another SIF extraction procedure is particularly relevant. Each of the eight COD fits

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(a) Turning-angle differences for linear COD fits.

(b) Turning-angle differences for quadratic COD fits.

(c) Mode-mixity differences for linear COD fits.

(d) Mode-mixity differences for quadratic COD fits.

Figure 6: COD–EDI differences along the late crack path for four normalized COD fitting windows and linear or quadratic extrapolation. Upper panels compare MTS turns; lower panels compare mode-mixity ratios. Zero denotes agreement of the extraction methods, not a known exact solution.

is 0.01024°, at P_{23} for the linear fit over $[0.12, 0.30]\Delta a$. The quadratic COD turns at that state range from 0.6204° to 0.6225°, compared with the EDI prediction of 0.62435°.

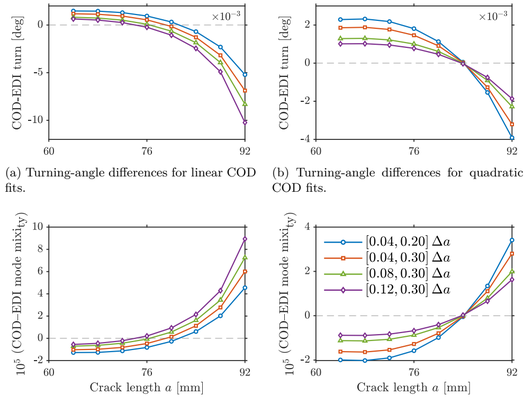
The distinction between the sign and magnitude of a small mode-mixity ratio is important close to local symmetry. At P_{21}, the rear-window linear fit exceeds the EDI ratio by 26.3%, although the corresponding turning-angle difference is only 0.00245°. The relative discrepancy is consequently sensitive to the small denominator in the comparison. The two extraction methods thus agree on the sign-change bracket, although their common displacement solution precludes an independent estimate of the finite-element error. The differences among COD fits measure sensitivity to the extrapolation procedure, not statistical uncertainty.

5.5 Sensitivity to structured-tip resolution

For comparisons at matched crack lengths, δx and δy denote coordinate differences between the compared and reference paths, $\delta r = (\delta x^2 + \delta y^2)^{1/2}$ is their separation, and $\delta\theta$ is the difference of absolute directions in the frozen frame. The symbol $\Delta\theta_{k+1}$ remains reserved for a local MTS turn. Differences in mode mixity and interpolated crossing length are denoted by δq_{K} and δa_{LS}, respectively.

Two aspects of the characteristic mode-mixity history are examined on fixed accepted crack geometries. At the positive maximum P_{17}, reducing the structured-tip spacing from 0.027012 to 0.013506 mm changes K_{I} by 0.00134%, q_{K} by 1.05×10^{-6}, and the MTS turn by -1.20×10^{-4} degrees. The small differences arise from alternative discretizations of the same physical geometry.

The late sign-change bracket provides a more sensitive test. Table 3 compares the P_{21} and P_{22} fields at three near-tip resolutions spanning a factor of four in h_{tip}. For every tested level, q_{K} remains positive at P_{21} and negative at P_{22}. The bracket and the corresponding change in MTS



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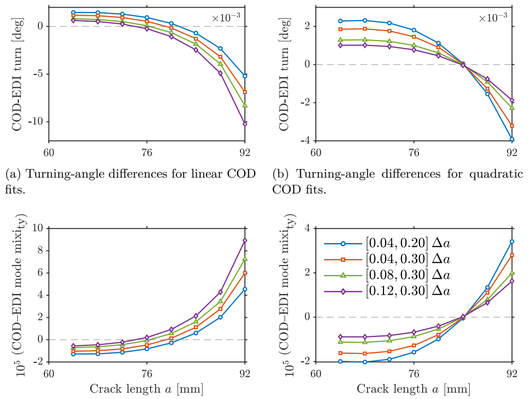
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gives $q_{\mathrm{K}} > 0$ at P_{21} and $q_{\mathrm{K}} < 0$ at P_{22}, in agreement with the EDI sign-change bracket. The quantitative differences depend on the chosen fitting window and polynomial degree (Fig. 6). Across 176 fits, the greatest absolute difference between the COD- and EDI-based MTS turns is 0.01024°, at P_{23} for the linear fit over $[0.12, 0.30]\Delta a$. The quadratic COD turns at that state range from 0.6204° to 0.6225°, compared with the EDI prediction of 0.62435°.

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Table 3: Structured-tip sensitivity at fixed P_{21} and P_{22} geometries, with common interaction-domain radii and requested exterior sizing. The reported a_{LS} values are interpolated sign-change estimates, not independently solved zero-mode-II states.

Resolution	h_{tip} [mm]	$q_{\mathrm{K}}(P_{21})$	$\Delta\theta_{22}$ [deg]	$q_{\mathrm{K}}(P_{22})$	$\Delta\theta_{23}$ [deg]	a_{LS} [mm]
$2h_0$	0.054025	$+7.7957\times10^{-5}$	−0.008933	-2.1451×10^{-3}	+0.245805	84.14027
h_0	0.027012	$+8.1343\times10^{-5}$	−0.009321	-2.1416×10^{-3}	+0.245406	84.14637
$h_0/2$	0.013506	$+8.3031\times10^{-5}$	−0.009515	-2.1399×10^{-3}	+0.245207	84.14941

turning sense thus persist under the tested near-tip refinement.

The linearly interpolated estimate a_{LS} varies from 84.14027 mm at $2h_0$ through 84.14637 mm at h_0 to 84.14941 mm at $h_0/2$. Its successive shifts of 6.10 and 3.04 μm give an indicative contraction ratio corresponding to an apparent order of approximately 1.00. This figure describes the mesh sensitivity of an *interpolated estimate*, not convergence of a solved zero-mode-II point. Because the contributing physical states are separated by a 4-mm crack increment, the small shifts cannot be interpreted as micrometer-scale accuracy of the local-symmetry location.

Propagation with the $2h_0$ tip mesh produces a path nearly indistinguishable from the reference trajectory at specimen scale [Fig. 7a]. The maximum vertical-coordinate difference is 0.12760 μm and the maximum Euclidean separation of corresponding tips is 0.12800 μm, both at P_{23}. The maximum absolute direction difference is 0.24539 millidegrees at P_{22} [Figs. 7b and 7c].

The two independently generated paths also retain the positive mode-mixity maximum at P_{17} and the sign-change bracket between P_{21} and P_{22} [Fig. 7d]. The largest difference in q_{K} is 2.65×10^{-7}. Linear interpolation of the two independently propagated P_{21}–P_{22} brackets gives $a_{\mathrm{LS}} = 84.14637$ mm for h_0 and 84.146276 mm for $2h_0$, a difference of −0.000094 mm (−0.094 μm). As above, this is only a sub-increment sensitivity diagnostic, not a micrometer-accurate physical location of local symmetry.

The positive mode-mixity maximum, sign-change bracket, and turning reversal persist across the tested structured-tip resolutions, both on fixed geometries and along the propagated $2h_0$ trajectory. This evidence concerns the stated mesh perturbations; the conforming exterior connectivity also changes when the core is remeshed.

5.6 Sensitivity to exterior-mesh density

Coarsening the exterior mesh has little effect on the SIFs of the fixed P_{17} and P_{22} geometries. We therefore examined its influence on the complete trajectory by propagating the M1 discretization from the common first segment P_0P_1. All later crack directions follow from the M1 solutions rather than from the reference path.

The two trajectories are nearly coincident at specimen scale [Fig. 8a]. Their small separation becomes apparent in the coordinate and angular differences in Fig. 8. The maximum vertical-coordinate difference is 0.06734 μm, the maximum Euclidean separation of corresponding tips is 0.06742 μm, and the maximum absolute direction difference is 0.07304 millidegrees, or 7.30×10^{-5} degrees [Figs. 8b and 8c].

The event sequence is also preserved when the exterior mesh is coarsened. Both propagated trajectories attain their positive mode-mixity maximum at P_{17} ($a = 68$ mm), with a positive ratio at P_{21} and a negative ratio at P_{22} [Fig. 8d]. Their largest absolute difference in q_{K} is 2.89×10^{-7}. Linear interpolation of the respective P_{21}–P_{22} brackets gives $a_{\mathrm{LS}} = 84.14637$ mm for the reference trajectory and 84.146 mm for M1, a difference of 0.000322 mm (0.322 μm). This sub-increment difference is reported only as a sensitivity diagnostic; neither interpolated value has physical precision at the micrometer scale.

The characteristic event sequence is therefore insensitive to the tested coarsening of the *exterior* mesh. Because the near-tip discretization and interaction-domain radii are unchanged, the

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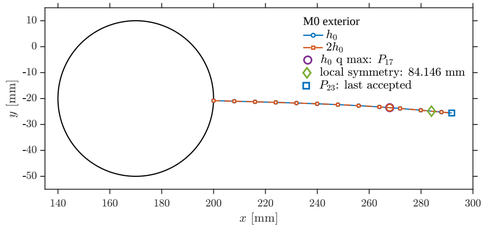
The two independently generated paths also retain the positive mode-mixity maximum at P_{17} and the sign-change bracket between P_{21} and P_{22} [Fig. 7d]. The largest difference in q_{K} is 2.65×10^{-7}. Linear interpolation of the two independently propagated P_{21}–P_{22} brackets gives $a_{\mathrm{LS}} = 84.14637$ mm for h_0 and 84.146276 mm for $2h_0$, a difference of −0.000094 mm (−0.094 μm). As above, this is only a sub-increment sensitivity diagnostic, not a micrometer-accurate physical location of local symmetry.

The positive mode-mixity maximum, sign-change bracket, and turning reversal persist across the tested structured-tip resolutions, both on fixed geometries and along the propagated $2h_0$ trajectory. This evidence concerns the stated mesh perturbations; the conforming exterior connectivity also changes when the core is remeshed.

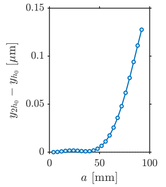
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Figure 7 captions and complete exterior-mesh discussion

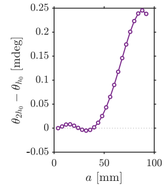
Author-supplied PDF (local stale Figure 8d) - page 16



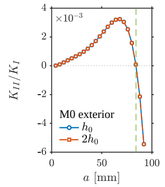
(a) Trajectories with structured-tip spacings h_0 and $2h_0$.



(b) Accumulated vertical tip-coordinate difference, $y_{2h_0} - y_{h_0}$.



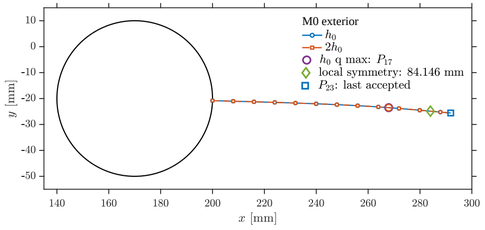
(c) Difference in the absolute crack direction in the frozen initiation frame, $\theta_{2h_0} - \theta_{h_0}$.



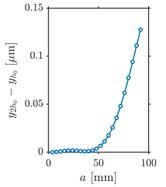
(d) Mode-mixity ratio K_{II}/K_I along the two independently propagated trajectories.

Figure 7: Effect of structured-tip resolution on the propagated trajectory. The h_0 and $2h_0$ paths share the interaction-domain radii and prescribed exterior sizing. The diamond indicates the interpolated mode-mixity zero of the reference path.

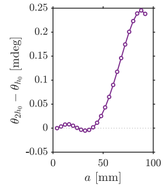
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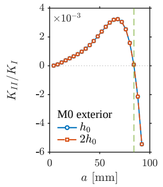
(a) Trajectories with structured-tip spacings h_0 and $2h_0$.



(b) Accumulated vertical tip-coordinate difference, $y_{2h_0} - y_{h_0}$.



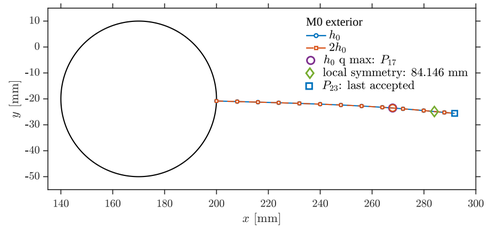
(c) Difference in the absolute crack direction in the frozen initiation frame, $\theta_{2h_0} - \theta_{h_0}$.



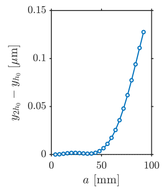
(d) Mode-mixity ratio K_{II}/K_I along the two independently propagated trajectories.

Figure 7: Effect of structured-tip resolution on the propagated trajectory. The h_0 and $2h_0$ paths share the interaction-domain radii and prescribed exterior sizing. The diamond indicates the interpolated mode-mixity zero of the reference path.

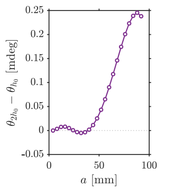
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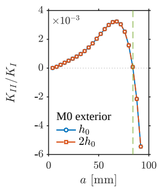
(a) Trajectories with structured-tip spacings h_0 and $2h_0$.



(b) Vertical tip-coordinate difference, $y_{2h_0} - y_{h_0}$.



(c) Crack-direction difference, $\theta_{2h_0} - \theta_{h_0}$.



(d) Mode-mixity histories for the two tip resolutions.

Figure 7: Effect of structured-tip resolution on the propagated trajectory. The h_0 and $2h_0$ paths share the interaction-domain radii and the M0 exterior sizing law. Panel (d) shows K_{II}/K_I. The diamond indicates the interpolated mode-mixity zero of the reference path.

5.6 Sensitivity to exterior-mesh density

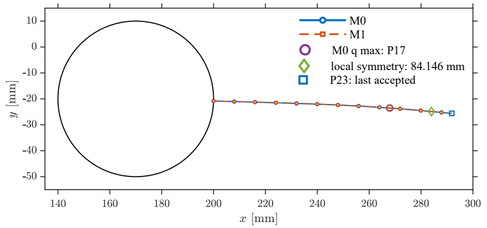
Coarsening the exterior mesh has little effect on the SIFs of the fixed P_{17} and P_{22} geometries. We therefore examined its influence on the complete trajectory by propagating the M1 discretization from the common first segment P_0P_1. All later crack directions follow from the M1 solutions rather than from the reference path.

The two trajectories are nearly coincident at specimen scale [Fig. 8a]. Their small separation becomes apparent in the coordinate and angular differences in Fig. 8. The maximum vertical-coordinate difference is 0.06734 μm, the maximum Euclidean separation of corresponding tips is 0.06742 μm, and the maximum absolute direction difference is 0.07304 millidegrees, or 7.30×10^{-5} degrees [Figs. 8b and 8c].

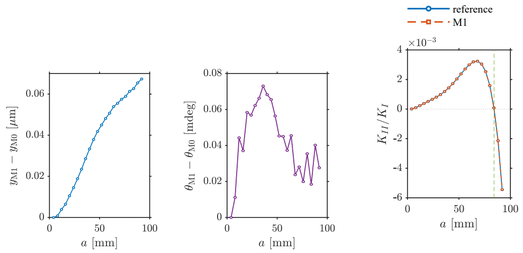
The event sequence is also preserved when the exterior mesh is coarsened. Both propagated trajectories attain their positive mode-mixity maximum at P_{17} ($a = 68$ mm), with a positive ratio at P_{21} and a negative ratio at P_{22} [Fig. 8d]. Their largest absolute difference in q_{K} is 2.89×10^{-7}. Linear interpolation of the respective P_{21}–P_{22} brackets gives $a_{\mathrm{LS}} = 84.14637$ mm for the reference trajectory and 84.146 mm for M1, a difference of 0.000322 mm (0.322 μm). This sub-increment difference is reported only as a sensitivity diagnostic; neither interpolated value has physical precision at the micrometer scale.

The characteristic event sequence is therefore insensitive to the tested coarsening of the *exterior* mesh. Because the near-tip discretization and interaction-domain radii are unchanged, the comparison does not establish general mesh independence.

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(a) Crack trajectories with the M0 and M1 exterior sizing laws.



(b) Accumulated vertical tip-coordinate difference, $y_{M1} - y_{M0}$.

(c) Difference in the absolute crack direction in the frozen initiation frame, $\theta_{M1} - \theta_{M0}$.

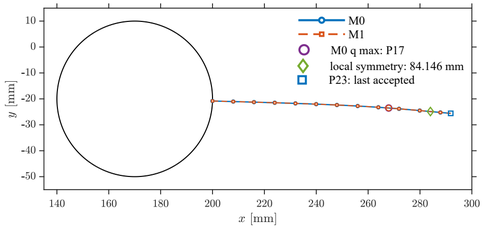
(d) Mode-mixity ratio K_{II}/K_I along the two independently propagated trajectories.

Figure 8: Effect of exterior-mesh coarsening on the crack path. The M0 and M1 trajectories are propagated separately with identical near-tip discretizations and interaction-domain radii. The diamond marks the interpolated mode-mixity zero of the reference path.

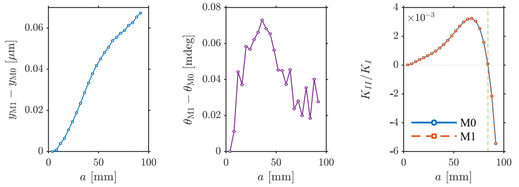
comparison does not establish general mesh independence.

5.7 Coupled crack-increment and discretization refinement

The paths computed with $\Delta a = 4$, 2, and 1 mm remain nearly coincident through 92 mm of growth [Fig. 9a]. Their differences in tip position and segment direction nevertheless decrease with refinement [Fig. 9]. In this study, the near-tip mesh, interaction domain, and exterior scales are reduced together with Δa, as specified in Section 4; hence the comparison concerns a coupled increment/discretization family.



(a) Crack trajectories with the M0 and M1 exterior sizing laws.



(b) Accumulated vertical tip-coordinate difference, $y_{M1} - y_{M0}$.

(c) Difference in the absolute crack direction in the frozen initiation frame, $\theta_{M1} - \theta_{M0}$.

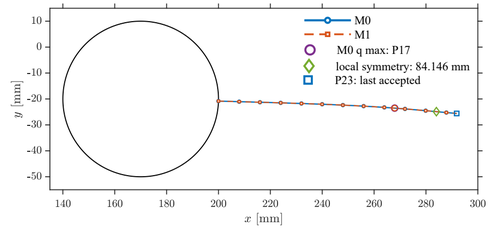
(d) Mode-mixity ratio K_{II}/K_I along the two independently propagated trajectories.

Figure 8: Effect of exterior-mesh coarsening on the crack path. The M0 and M1 trajectories are propagated separately with identical near-tip discretizations and interaction-domain radii. The diamond marks the interpolated mode-mixity zero of the reference path.

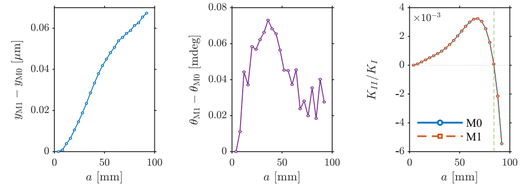
comparison does not establish general mesh independence.

5.7 Coupled crack-increment and discretization refinement

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(a) Crack trajectories with the M0 and M1 exterior sizing laws.



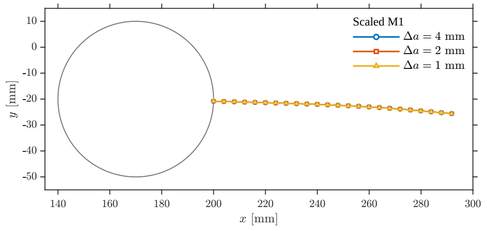
(b) Vertical tip-coordinate difference, $y_{M1} - y_{M0}$.

(c) Crack-direction difference, $\theta_{M1} - \theta_{M0}$.

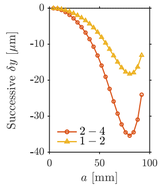
(d) Mode-mixity histories for M0 and M1.

Figure 8: Effect of exterior-mesh coarsening on the crack path. The M0 and M1 trajectories are propagated separately with identical near-tip discretizations and interaction-domain radii. Panel (d) shows K_{II}/K_I. The diamond marks the interpolated mode-mixity zero of the reference path.

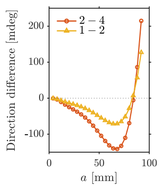
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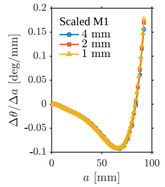
(a) Independently propagated crack trajectories for $\Delta a = 4$, 2, and 1 mm, all continued to 92 mm.



(b) Successive vertical-coordinate differences, $y_{2\,\mathrm{mm}} - y_{4\,\mathrm{mm}}$ and $y_{1\,\mathrm{mm}} - y_{2\,\mathrm{mm}}$, evaluated at exact common crack lengths.

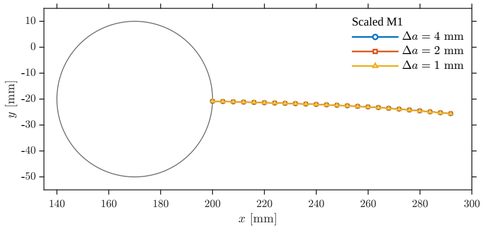


(c) Successive differences in absolute crack direction in the frozen initiation frame.

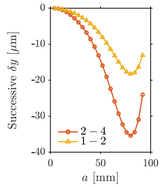


(d) Turning density $\Delta\theta/\Delta a$ along the three independently propagated trajectories.

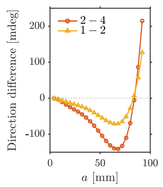
Figure 9: Coupled crack-increment and discretization refinement. The 4-, 2-, and 1-mm paths share dimensionless mesh and interaction-domain scales. Panels (b,c) compare paths at common crack lengths; panel (d) shows turning density on each path's native grid.



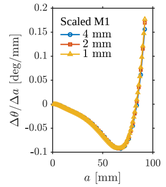
(a) Independently propagated crack trajectories for $\Delta a = 4$, 2, and 1 mm, all continued to 92 mm.



(b) Successive vertical-coordinate differences, $y_{2\,\mathrm{mm}} - y_{4\,\mathrm{mm}}$ and $y_{1\,\mathrm{mm}} - y_{2\,\mathrm{mm}}$, evaluated at exact common crack lengths.



(c) Successive differences in absolute crack direction in the frozen initiation frame.

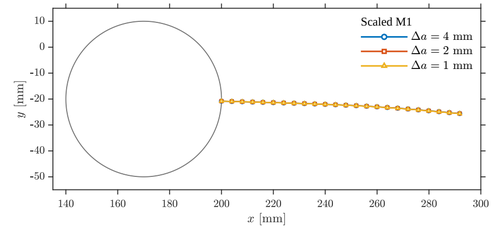


(d) Turning density $\Delta\theta/\Delta a$ along the three independently propagated trajectories.

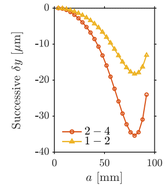
Figure 9: Coupled crack-increment and discretization refinement. The 4-, 2-, and 1-mm paths share dimensionless mesh and interaction-domain scales. Panels (b,c) compare paths at common crack lengths; panel (d) shows turning density on each path's native grid.

5.7 Coupled crack-increment and discretization refinement

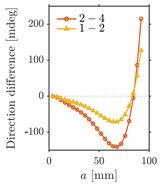
The paths computed with $\Delta a = 4$, 2, and 1 mm remain nearly coincident through 92 mm of growth [Fig. 9a]. Their differences in tip position and segment direction nevertheless decrease with refinement [Fig. 9]. In this study, the near-tip mesh, interaction domain, and exterior scales are reduced together with Δa, as specified in Section 4; hence the comparison concerns a coupled increment/discretization family.



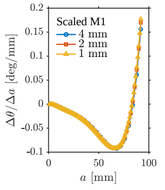
(a) Independently propagated crack trajectories for $\Delta a = 4$, 2, and 1 mm, all continued to 92 mm.



(b) Successive vertical tip-coordinate differences.



(c) Successive crack-direction differences.



(d) Turning density along the three trajectories.

Figure 9: Coupled crack-increment and discretization refinement. The 4-, 2-, and 1-mm paths share dimensionless mesh and interaction-domain scales within the scaled-M1 family. Panel (b) shows $y_{2\,\mathrm{mm}} - y_{4\,\mathrm{mm}}$ and $y_{1\,\mathrm{mm}} - y_{2\,\mathrm{mm}}$ at exact common crack lengths. Panel (c) shows the corresponding differences of absolute directions in the frozen initiation frame. Panel (d) shows turning density $\Delta\theta/\Delta a$ on each path's native grid.

Table 4 compares tip positions and directions at crack lengths common to all three paths. The maximum tip separation contracts from 35.472 μm for 4 versus 2 mm to 18.291 μm for 2 versus 1 mm. The corresponding vertical differences decrease from 35.418 to 18.263 μm, while the maximum absolute direction difference decreases from 214.931 to 127.391 millidegrees.

Table 4: Successive differences between independently propagated paths in the dimensionless crack-increment refinement family. Comparisons use exact common crack lengths; no path interpolation is used.

Comparison	$\max\lvert\delta y\rvert$ [μm]	$\max\delta r$ [μm]	$\max\lvert\delta\theta\rvert$ [mdeg]	$\max\lvert\delta q_{\mathrm{K}}\rvert$
2 mm − 4 mm	35.4181	35.4721	214.931	2.48578×10^{-3}
1 mm − 2 mm	18.2629	18.2911	127.391	1.41618×10^{-3}

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Table 4 compares tip positions and directions at crack lengths common to all three paths. The maximum tip separation contracts from 35.472 μm for 4 versus 2 mm to 18.291 μm for 2 versus 1 mm. The corresponding vertical differences decrease from 35.418 to 18.263 μm, while the maximum absolute direction difference decreases from 214.931 to 127.391 millidegrees.

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2 mm − 4 mm	35.4181	35.4721	214.931	2.48578×10^{-3}
1 mm − 2 mm	18.2629	18.2911	127.391	1.41618×10^{-3}

The contraction ratios are 1.939 for maximum tip separation and 1.687 for maximum direction difference. If d denotes a successive-difference maximum, the descriptive quantity $p = \log_2(d_{\Delta a}/d_{\Delta a/2})$ takes values of approximately 0.96 and 0.75, respectively. These are not formal Richardson orders: the compared paths result from recursively updated geometries, the maximizers need not occur at the same crack length, and both the increment and discretization scales change. These values describe contraction within the three-level family and cannot be regarded as measured asymptotic rates.

The normalized mode-mixity response exhibits a complementary trend. As shown in Table 5, the positive maximum of q_{K} falls from 3.23937×10^{-3} at 4 mm to 1.61266×10^{-3} at 2 mm and 8.04068×10^{-4} at 1 mm. Despite this nearly proportional decrease, the maximum divided by Δa changes by only about 0.71% between the coarsest and finest levels. The corresponding MTS turning density ranges from -0.092799 to -0.092139 deg/mm. The turning-density histories in Fig. 9d exhibit similar shapes on their respective crack-length grids, although this does not establish convergence of a smooth limiting curvature.

The positive maximum is sampled at $a = 68$ mm on the 4-mm path and at $a = 66$ mm on the two finer paths; the values therefore do not correspond to an identical crack length. Nevertheless, their near-proportional decrease with Δa, together with the stability of turning density, is consistent with the conditional curvature–MTS correspondence in Eq. (17). The observed behavior may be summarized as

$$q_{\mathrm{K}} = O(\Delta a), \qquad \Delta\theta = O(\Delta a), \tag{28}$$

within the investigated numerical family. Three coupled refinement levels provide evidence of this scaling rather than an independent proof of asymptotic convergence.

The estimated position of the mode-mixity sign change also shifts less with successive refinement. Linear interpolation gives $a_{\mathrm{LS}} = 84.148913$, 83.550120, and 83.268112 mm for the three levels. The successive differences are 0.59879 and 0.28201 mm, giving a contraction ratio of 2.12 and a descriptive apparent order near 1.09. On the 1-mm path, the sign change is bracketed by the solved states at 83 and 84 mm. None of these interpolated estimates is itself a solved zero-K_{II} state.

The nearly proportional decrease of the raw mode-mixity maxima and the stability of the increment-normalized measures accord with the conditional MTS–curvature relation in Eq. (17).

Table 5: Characteristic quantities in the crack-increment refinement family. The local-symmetry location is obtained by linear interpolation between the two solved states bracketing $q_{\mathrm{K}} = 0$.

Δa [mm]	$a(q_{\mathrm{K,max}})$ [mm]	$q_{\mathrm{K,max}}$	$q_{\mathrm{K,max}}/\Delta a$ [mm^{-1}]	$(\Delta\theta/\Delta a)_{q_{\max}}$ [deg/mm]	a_{LS} [mm]
4	68	3.2394×10^{-3}	8.0984×10^{-4}	−0.092799	84.148913
2	66	1.6127×10^{-3}	8.0633×10^{-4}	−0.092398	83.550120
1	66	8.0407×10^{-4}	8.0407×10^{-4}	−0.092139	83.268112

Table 4 compares tip positions and directions at crack lengths common to all three paths. The maximum tip separation contracts from 35.472 μm for 4 versus 2 mm to 18.291 μm for 2 versus 1 mm. The corresponding vertical differences decrease from 35.418 to 18.263 μm, while the maximum absolute direction difference decreases from 214.931 to 127.391 millidegrees.

Table 4: Successive differences between independently propagated paths in the dimensionless crack-increment refinement family. Comparisons use exact common crack lengths; no path interpolation is used.

Comparison	max $\lvert\delta y\rvert$ [μm]	max δr [μm]	max $\lvert\delta\theta\rvert$ [mdeg]	max $\lvert\delta q_{\mathrm{K}}\rvert$
2 mm − 4 mm	35.4181	35.4721	214.931	2.48578×10^{-3}
1 mm − 2 mm	18.2629	18.2911	127.391	1.41618×10^{-3}

The contraction ratios are 1.939 for maximum tip separation and 1.687 for maximum direction difference. If d denotes a successive-difference maximum, the descriptive quantity $p = \log_2(d_{\Delta a}/d_{\Delta a/2})$ takes values of approximately 0.96 and 0.75, respectively. These are not formal Richardson orders: the compared paths result from recursively updated geometries, the maximizers need not occur at the same crack length, and both the increment and discretization scales change. These values describe contraction within the three-level family and cannot be regarded as measured asymptotic rates.

The normalized mode-mixity response exhibits a complementary trend. As shown in Table 5, the positive maximum of q_{K} falls from 3.23937×10^{-3} at 4 mm to 1.61266×10^{-3} at 2 mm and 8.04068×10^{-4} at 1 mm. Despite this nearly proportional decrease, the maximum divided by Δa changes by only about 0.71% between the coarsest and finest levels. The corresponding MTS turning density ranges from -0.092799 to -0.092139 deg/mm. The turning-density histories in Fig. 9d exhibit similar shapes on their respective crack-length grids, although this does not establish convergence of a smooth limiting curvature.

The positive maximum is sampled at $a = 68$ mm on the 4-mm path and at $a = 66$ mm on the two finer paths; the values therefore do not correspond to an identical crack length. Nevertheless, their near-proportional decrease with Δa, together with the stability of turning density, is consistent with the conditional curvature–MTS correspondence in Eq. (17). The observed behavior may be summarized as

$$q_{\mathrm{K}} = O(\Delta a), \qquad \Delta\theta = O(\Delta a), \tag{28}$$

within the investigated numerical family. Three coupled refinement levels provide evidence of this scaling rather than an independent proof of asymptotic convergence.

The estimated position of the mode-mixity sign change also shifts less with successive refinement. Linear interpolation gives $a_{\mathrm{LS}} = 84.148913$, 83.550120, and 83.268112 mm for the three levels. The successive differences are 0.59879 and 0.28201 mm, giving a contraction ratio of 2.12 and a descriptive apparent order near 1.09. On the 1-mm path, the sign change is bracketed by the solved states at 83 and 84 mm. None of these interpolated estimates is itself a solved zero-K_{II} state.

The nearly proportional decrease of the raw mode-mixity maxima and the stability of the increment-normalized measures accord with the conditional MTS–curvature relation in Eq. (17).

Table 5: Characteristic quantities in the crack-increment refinement family. The local-symmetry location is obtained by linear interpolation between the two solved states bracketing $q_{\mathrm{K}} = 0$.

Δa [mm]	$a(q_{\mathrm{K,max}})$ [mm]	$q_{\mathrm{K,max}}$	$q_{\mathrm{K,max}}/\Delta a$ [mm^{-1}]	$(\Delta\theta/\Delta a)_{q_{\max}}$ [deg/mm]	a_{LS} [mm]
4	68	3.2394×10^{-3}	8.0984×10^{-4}	−0.092799	84.148913
2	66	1.6127×10^{-3}	8.0633×10^{-4}	−0.092398	83.550120
1	66	8.0407×10^{-4}	8.0407×10^{-4}	−0.092139	83.268112

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6 Discussion

6.1 Mechanical interpretation of the turning reversal

The turning sense of each new segment follows from the current-tip mode-mixity ratio through the MTS criterion. Positive q_{K} produces a clockwise correction under the adopted sign convention; when q_{K} becomes negative, the correction is counterclockwise. The positive mode-mixity maximum at P_{17}, followed by the sign change between P_{21} and P_{22}, therefore explains the changes in turning direction along the accepted trajectory. At the same time, K_{I} increases under the fixed unit remote traction. The reversal is consequently a change in the relative sliding-mode contribution, not a decrease in the opening-mode intensity or a reversal of the crack's vertical displacement.

The remaining horizontal ligament is only 8.16 mm at P_{23}, suggesting that the approaching traction-free right boundary may contribute to the late mixed-mode response. That influence cannot, however, be identified separately from the increasing crack length and the evolving crack geometry in the present calculations. A controlled comparison at different boundary distances would be needed to determine its contribution. The computed SIFs establish the MTS turning mechanism, but not a unique physical cause for their evolution.

6.2 Numerical sensitivity and the smooth-path interpretation

The verification studies address different sources of uncertainty. Prescribed asymptotic fields test the implementation of SIF extraction, whereas COD fitting and the interaction integral provide two estimates derived from the same finite-element displacement field. Their agreement near the mode-mixity sign change lends support to the extracted turning sense, but cannot serve as an independent validation of the underlying elastic solution. Likewise, the centered-hole test establishes the expected reflection symmetry and a small one-step angular response, without implying attraction of an entire evolving trajectory.

The independently propagated mesh studies examine whether discretization differences alter the recursive path construction. Coarsening the exterior mesh in M1 preserves the paired near-tip core and interaction-domain support; changing the structured-tip spacing preserves the requested exterior sizing law, though conforming remeshing can change the exterior connectivity. Both propagated alternatives retain the positive mode-mixity maximum and the P_{21}–P_{22} sign-change bracket. Their maximum corresponding-tip separations, 0.0674 and 0.128 μm, respectively, are small relative to the crack increments. These observations demonstrate low sensitivity to the two tested mesh perturbations, not independence of the solution from all discretization choices.

The 4/2/1-mm trajectories probe a different aspect of the algorithm. In that family, the crack increment, structured-tip spacing, interaction-domain radii, and exterior mesh scales decrease together. Successive differences in tip position and direction contract, while the positive maximum of q_{K} decreases approximately in proportion to Δa. Its increment-normalized value and the corresponding turning density remain comparatively stable. The small-mixity MTS expansion thus provides a plausible connection to the signed curvature of a smooth limiting trajectory [Eq. (17)], provided such a limit exists and the discrete turning densities converge. Neither a formal convergence order nor the separate contribution of crack-increment error follows from these three coupled refinement levels. This distinction also limits direct comparison with studies that vary the increment alone, including the two-level comparison of Fang et al. [11].

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Coarser exterior M1	0.06734	0.06742	0.07304	2.89×10^{-7}	-0.322
Isolated-core $2h_0$	0.12760	0.12800	0.24539	2.65×10^{-7}	-0.094

Further tests should vary the interaction-domain radii while holding the physical crack geometry and other numerical parameters fixed. The existing mesh-quality, residual, and prescribed-field checks establish admissibility of the numerical calculations, but they do not provide a comprehensive error bound on the physical SIFs or trajectory. The earlier 8-mm diagnostic is retained as an extraction test and is not part of the coupled refinement sequence.

6.3 Last accepted state and physical scope

The accepted crack history ends at P_{23}. The proposed $P_{23}P_{24}$ extension satisfies geometrical and prescribed-field checks, but the attempted elastic solution for the 96-mm crack did not meet the PCG residual criterion. The resulting numerical stopping point cannot be interpreted as fracture arrest or contact with the right boundary.

The model treats one active crack in a homogeneous, linearly elastic plate and supplies only the directions of imposed quasi-static extensions. The stress-based initiation rule is not a subsequent propagation threshold, and the calculation does not determine growth loads, rates, or arrest. Nor does it model branching, competing initiation, plasticity, or dynamic fracture. These processes remain outside the formulation rather than being ruled out by the computed path.

7 Conclusions

An incremental LEFM procedure has been used to obtain the trajectory of a single crack emanating from an asymmetrically located circular hole. The uncracked-hole stress field selects the initiation point, and a prescribed outward-normal first segment establishes the initial crack geometry. All subsequent directions follow from current-tip interaction-integral SIFs and the MTS criterion, with the entire previous path retained in each elastic problem. The accepted 4-mm history extends to 92 mm; the next qualified geometry has no accepted physical solution.

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The numerical comparisons support the persistence of these characteristic events under the tested near-tip and exterior-mesh changes. A centered-hole control verifies the one-step response under reflection symmetry but does not establish stability of the evolving path. In the independently propagated 4/2/1-mm family, the geometric and directional differences contract as the increment and numerical length scales decrease together. The nearly proportional reduction in raw mode mixity and the relative stability of increment-normalized turning measures are consistent with a conditional smooth-path interpretation, not a proven continuum-convergence result.

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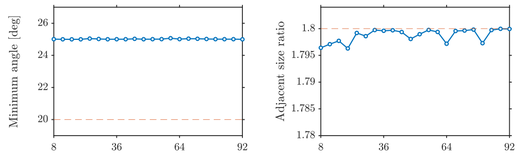
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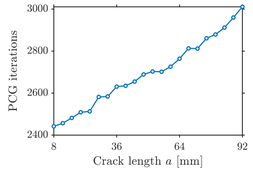
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The accepted solver and mesh checks are summarized in Fig. S1. They provide reproducibility and acceptance diagnostics, rather than an estimate of the discretization error in the trajectory.

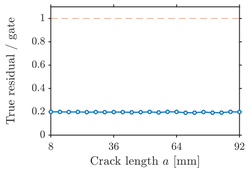


(a) Minimum triangle-angle check.

(b) Adjacent element-size-ratio check.



(c) PCG iterations for accepted physical solves.



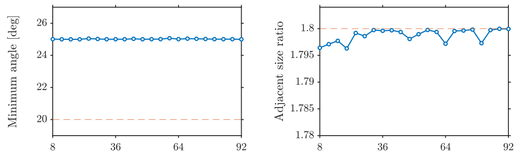
(d) True relative residual normalized by its acceptance gate.

Figure S1: Diagnostics for accepted P_2–P_{23} states. Dashed lines are the 20° minimum-angle gate, 1.8 adjacent-size-ratio gate, and normalized true-residual gate of one. The separately enforced reported-PCG tolerance is 10^{-10}; the true-residual acceptance denominator is 5×10^{-10}. No failed P_{24} solve is plotted.

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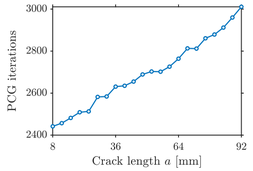
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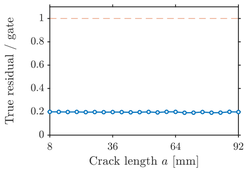


(a) Minimum triangle-angle check.

(b) Adjacent element-size-ratio check.



(c) PCG iterations for accepted physical solves.



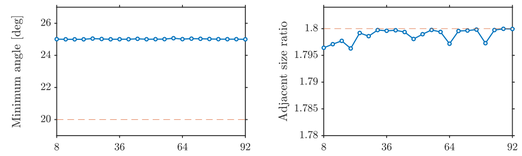
(d) True relative residual normalized by its acceptance gate.

Figure S1: Diagnostics for accepted P_2–P_{23} states. Dashed lines are the 20° minimum-angle gate, 1.8 adjacent-size-ratio gate, and normalized true-residual gate of one. The separately enforced reported-PCG tolerance is 10^{-10}; the true-residual acceptance denominator is 5×10^{-10}. No failed P_{24} solve is plotted.

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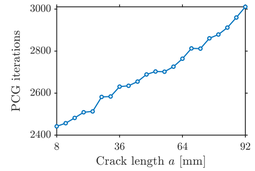
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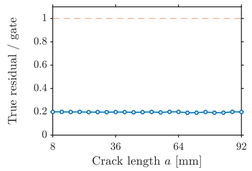


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